Name: _____ Date: _____ Period: _____

3-3 (Day 2): Divide First, Then Derive It

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Tags: B = Basic, I = Intermediate, A = Advanced.

Warm-up: Day 1 skills

W1. (B) Solve by completing the square: $\mathbf{x^2 - 6x - 3 = 0}$

W2. (B) Fill in: $x^2 + 5x + _____ = (\ x + _____\ )^2$

Part A: When the leading coefficient is not 1

New first move: **divide every term (both sides) by a**, so the x^2 coefficient becomes 1. Then do what you did on day 1.

1. (I) Solve $\mathbf{3x^2 + 12x - 9 = 0}$. Fill in the steps.

Step 1. Divide every term by _____ : $x^2 +$ _____ $x -$ _____ $= 0$

Step 2. Move the constant: $x^2 +$ _____ $x =$ _____

Step 3. Add (half of the x-coefficient)$^2 = (\ $_____ $)^2 =$ _____ to both sides: $x^2 +$ _____ $x +$ _____ $=$ _____

Step 4. Factor the left side: $(\ x +$ _____ $)^2 =$ _____

Step 5. Square root both sides: $x +$ _____ $= \pm \sqrt{\ }$ _____

Step 6. Solve for x: $x =$ _____ $\pm \sqrt{\ }$ _____ (Check one answer on your calculator.)

2. (I) Solve $\mathbf{4x^2 - 8x - 5 = 0}$. Give exact answers.

3. (I) Solve $\mathbf{2x^2 + 5x - 1 = 0}$. Give exact answers, then decimals to the nearest hundredth.

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Part A, continued. Give exact answers. Check one solution by substituting.

4. (I) Solve $5x^2 + 10x + 10 = 0$. (Your answers may be complex.)

5. (I) Solve $-x^2 + 4x + 1 = 0$.

6. (I) Solve $\frac{1}{2}x^2 - 3x + 1 = 0$. (What do you divide by? Careful with fractions.)

Part B: Derive the quadratic formula (A)

Left: the moves done on numbers. Right: the **same moves** on letters. Fill in every blank on the right ($a \neq 0$).

Move	**Numbers: $2x^2 + 3x - 5 = 0$**	**Letters: $ax^2 + bx + c = 0$**
1. Divide every term by a	$x^2 + (3/2)x - 5/2 = 0$	$x^2 +$ _____ $x +$ _____ $= 0$
2. Move the constant	$x^2 + (3/2)x = 5/2$	$x^2 + (b/a)x =$ _____
3. Add (half the x-coefficient)^2 to both sides	$(3/4)^2 = 9/16$ $x^2 + (3/2)x + 9/16 = 5/2 + 9/16$	Half of b/a is _____; squared: _____ $x^2 + (b/a)x +$ _____ $= -c/a +$ _____
4. Factor left, combine right	$(x + 3/4)^2 = 40/16 + 9/16 = 49/16$	$(\ x +$ _____ $)^2 =$ _____ $/\ 4a^2$ (common denominator $4a^2$)
5. Square roots	$x + 3/4 = \pm\ 7/4$	$x +$ _____ $= \pm\ \sqrt{}$ _____ $/$ _____
6. Solve for x	$x = (-3 \pm 7)/4 = 1$ or $-5/2$	$x =$ _____ (the quadratic formula)

In move 3, why do we add $(b/2a)^2$ and not $(b/2)^2$? (Hint: after move 1, what number is in front of x?)

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Part C: Use the formula

For each equation: write a, b, c (with signs), find $b^2 - 4ac$, predict the type (*two real*, *one repeated real*, or *two complex*), then solve.

#	Equation	a, b, c	$b^2 - 4ac$	Predict	Solutions (exact)
7 (I)	$3x^2 - 7x + 1 = 0$				
8 (I)	$6x^2 + x - 2 = 0$				
9 (I)	$4x^2 - 20x + 25 = 0$				
10 (I)	$2x^2 - 2x + 5 = 0$				

11. (A) For $\mathbf{x^2 + 6x + k = 0}$, which values of k give two real solutions? One repeated real solution? Two complex solutions? (Hint: complete the square: $(x + 3)^2 =$ _____.)

12. (I) (A) A ball is thrown up from a roof. Its height in feet after t seconds is $\mathbf{h(t) = -16t^2 + 40t + 6}$.

(a) When is the ball 20 feet above the ground? Round to the nearest hundredth of a second.

(b) Does the ball ever reach 40 feet? Use the discriminant, and explain in words.

Math Time Machine: Brahmagupta (India, 628 CE) wrote a rule in *words* that matches the quadratic formula, among the earliest known. In 1545 Cardano split 10 into two parts with product 40 and got $5 \pm \sqrt{-15}$, numbers he found "sophisticated" but used anyway. Euler gave $\sqrt{-1}$ the name i in 1777.